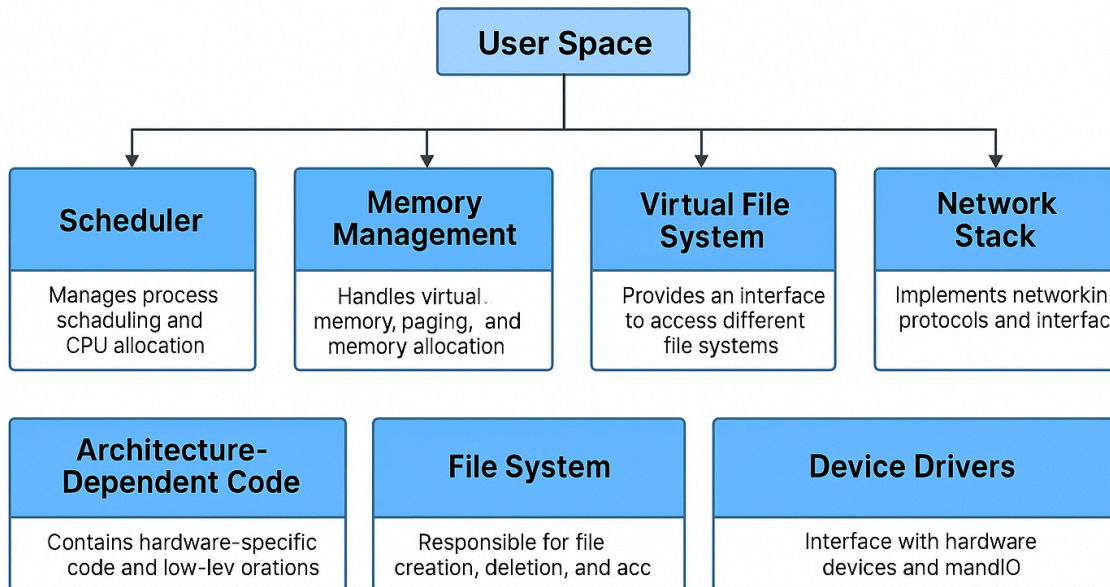


Linux Kernel Architecture



Descriptions of Major Components:

1. Process Scheduler:

- Manages process execution by determining which process runs at a given time.
- It ensures fair CPU usage and handles multitasking by switching between processes efficiently.

2. Memory Management:

- Manages physical and virtual memory, paging, and segmentation.
- It allocates memory to processes and ensures isolation and protection.

3. Virtual File System (VFS):

- Provides a uniform interface to access different types of file systems.
- It enables user programs to interact with files and directories regardless of the underlying storage format.

4. Network Stack:

- Handles network communication using protocols like TCP/IP.
- Manages sockets, packet routing, and network interfaces.

5. Device Drivers:

- Allow the kernel to communicate with hardware devices like disks, keyboards, and printers.
- They translate generic kernel commands into device-specific operations.

6. Interprocess Communication (IPC):

- Provides mechanisms like pipes, message queues, and shared memory.
- Enables processes to communicate and synchronize with each other.

7. System Call Interface:

- Acts as a bridge between user space and kernel space.
- Provides functions that user applications use to request kernel services.

8. Hardware Abstraction Layer (HAL):

- Hides hardware specifics and provides a consistent interface to the kernel.
- Ensures portability across different hardware platforms.